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Method for manufacturing light detection device, light detection device, and electronic device

Abstract

Provided is a method for manufacturing a light detection device capable of suppressing optical color mixing while improving sensitivity. On a substrate in which a plurality of photoelectric conversion units is formed, anisotropic etching is performed from one surface side of the substrate to form a plurality of openings arranged in a lattice shape at predetermined intervals on the substrate to surround each photoelectric conversion unit. Subsequently, isotropic etching is performed from one surface side of the substrate to connect adjacent openings to each other, to form a lattice-shaped trench part on the substrate to surround each photoelectric conversion unit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2021/021233 filed on Jun. 3, 2021, which claims priority benefit of Japanese Patent Application No. JP 2020-100844 filed in the Japan Patent Office on Jun. 10, 2020. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present technology relates to a method for manufacturing a light detection device, a light detection device, and an electronic device.

BACKGROUND ART

(3) Conventionally, there has been proposed a light detection device including a substrate, a plurality of photoelectric conversion units formed in the substrate, and a lattice-shaped trench part formed on the substrate to surround each photoelectric conversion unit, in which an insulating film or metal is embedded in the trench part to reduce optical color mixing (see, for example, Patent Document 1).

(4) In such a light detection device, the trench part is usually formed by performing etching using an etching mask having a lattice-shaped opening corresponding to a shape of the trench part.

CITATION LIST

Patent Document

(5) Patent Document 1: Japanese Patent Application Laid-Open No. 2013-175494

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(6) However, in a case where etching is performed using the etching mask having a lattice-shaped opening at the time of forming the trench part, there is a possibility that a corner part of an intersection part of the trench part is rounded and an area of the corner part is reduced. Therefore, there is a possibility that vignetting of incident light occurs at the intersection part, and sensitivity is reduced.

(7) An object of the present disclosure is to provide a method for manufacturing a light detection device capable of suppressing optical color mixing while improving sensitivity, a light detection device, and an electronic device.

Solutions to Problems

(8) A method for manufacturing a light detection device according to the present disclosure includes: (a) performing, on a substrate in which a plurality of photoelectric conversion units is formed, anisotropic etching from one surface side of the substrate to form a plurality of openings arranged in a lattice shape at predetermined intervals on the substrate to surround each photoelectric conversion unit; and (b) performing isotropic etching from one surface side of the substrate to connect the openings adjacent to each other, to form a lattice-shaped trench part on the substrate to surround each photoelectric conversion unit.

(9) Furthermore, a light detection device according to the present disclosure includes: (a) a substrate; (b) a plurality of photoelectric conversion units formed in the substrate; and (c) a lattice-shaped trench part formed on the substrate to surround each photoelectric conversion unit, in which (d) a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a longitudinal direction of a linear trench part constituting the trench part.

(10) Furthermore, an electronic device of the present disclosure includes: (a) a light detection device including a substrate, a plurality of photoelectric conversion units formed in the substrate, and a lattice-shaped trench part formed on the substrate to surround each photoelectric conversion

unit, in which a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a longitudinal direction of a linear trench part constituting the trench part; (b) an optical lens configured to form an image of image light from a subject, on an imaging surface of the light detection device; and (c) a signal processing circuit configured to perform signal processing on a signal outputted from the light detection device.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a diagram illustrating an overall configuration of a solid-state imaging device according to a first embodiment.
- (2) FIG. 2 is a view illustrating a cross-sectional configuration of a pixel region in a case of being cut along line A-A in FIG. 1.
- (3) FIG. 3 is a view illustrating a planar configuration of the pixel region in a case of being cut along line B-B in FIG. 2.
- (4) FIG. 4A is an enlarged view illustrating a planar configuration of an intersection part of a trench part illustrated in FIG. 3.
- (5) FIG. 4B is a view illustrating a cross-sectional configuration of a substrate in a case of being cut along line C-C in FIG. 4A.
- (6) FIG. 5A is a view illustrating each step of a method for manufacturing a solid-state imaging device according to the first embodiment.
- (7) FIG. 5B is a view illustrating each step of the method for manufacturing the solid-state imaging device according to the first embodiment.
- (8) FIG. 6A is a view illustrating a planar configuration of an etching mask formed on a back surface of the substrate.
- (9) FIG. 6B is a view illustrating a cross-sectional configuration of the substrate in a case of being cut along line D-D in FIG. 6A.
- (10) FIG. 7 is a view illustrating a photomask for forming a resist mask illustrated in FIG. 6A.
- (11) FIG. 8A is a view illustrating a cross-sectional configuration of the substrate after anisotropic etching.
- (12) FIG. 8B is a view illustrating a cross-sectional configuration of the substrate in a case of being cut along line E-E in FIG. 8A.
- (13) FIG. 9 is a view illustrating a cross-sectional configuration of the substrate after isotropic etching.
- (14) FIG. 10 is a view illustrating a width of a trench part obtained by the method for manufacturing the solid-state imaging device according to the first embodiment.
- (15) FIG. 11A is a view illustrating a cross-sectional configuration of a pixel region of a solid-state imaging device according to a modification.
- (16) FIG. 11B is a view illustrating a step of forming the trench part illustrated in FIG. 11A.
- (17) FIG. 12A is a view illustrating a planar configuration of an etching mask according to a modification.
- (18) FIG. 12B is an enlarged view illustrating a planar configuration of an intersection part of a trench part formed by the etching mask illustrated in FIG. 12A.
- (19) FIG. 13A is a view illustrating a planar configuration of an etching mask according to a modification.
- (20) FIG. 13B is an enlarged view illustrating a planar configuration of an intersection part of a trench part formed by the etching mask illustrated in FIG. 13A.
- (21) FIG. 14A is a view illustrating a planar configuration of an etching mask according to a

modification.

(22) FIG. **14B** is an enlarged view illustrating a planar configuration of an intersection part of a trench part formed by the etching mask illustrated in FIG. **14A**.

(23) FIG. **15** is a view illustrating a photomask for forming a resist mask illustrated in FIG. **12A**.

(24) FIG. **16** is a view illustrating a cross-sectional configuration of a pixel region of a solid-state imaging device according to a modification.

(25) FIG. **17** is a view illustrating a cross-sectional configuration of a pixel region of a solid-state imaging device according to a modification.

(26) FIG. **18** is a view illustrating a flow of a manufacturing step for a solid-state imaging device according to a modification.

(27) FIG. **19** is a view illustrating a cross-sectional configuration of a pixel region of a solid-state imaging device according to a modification.

(28) FIG. **20** is a view illustrating an overall configuration of a chip in which a solid-state imaging device according to a second embodiment is formed.

(29) FIG. **21A** is a view illustrating a planar configuration of the chip in a case where a region F in FIG. **20** is enlarged.

(30) FIG. **21B** is a view illustrating a cross-sectional configuration of the chip in a case of being cut along line G-G in FIG. **21A**.

(31) FIG. **22A** is an enlarged view illustrating a planar configuration of a groove illustrated in FIG. **21A**.

(32) FIG. **22B** is a view illustrating a cross-sectional configuration of a substrate in a case of being cut along line H-H in FIG. **22A**.

(33) FIG. **23** is a view illustrating a planar configuration of an etching mask formed on a back surface of the substrate.

(34) FIG. **24** is a view illustrating a cross-sectional configuration of the substrate in a case of being cut along line I-I in FIG. **23**.

(35) FIG. **25** is a view illustrating a photomask for forming a resist mask illustrated in FIG. **23**.

(36) FIG. **26A** is a view illustrating a cross-sectional configuration of the substrate after anisotropic etching.

(37) FIG. **26B** is a view illustrating a cross-sectional configuration of the substrate in a case of being cut along line J-J in FIG. **26A**.

(38) FIG. **27** is a view illustrating a cross-sectional configuration of the substrate after isotropic etching.

(39) FIG. **28** is a view illustrating a width of a groove obtained by a method for manufacturing a solid-state imaging device according to the second embodiment.

(40) FIG. **29** is a view illustrating a planar configuration of the chip in which one of side wall surfaces of the groove has a sawtooth shape.

(41) FIG. **30** is a view illustrating a planar configuration of an etching mask used in a manufacturing step for a solid-state imaging device according to a modification.

(42) FIG. **31A** is a view illustrating a cross-sectional configuration of the substrate in a case of being cut along line K-K in FIG. **30**.

(43) FIG. **31B** is a view illustrating a cross-sectional configuration of the substrate after isotropic etching.

(44) FIG. **32A** is a view illustrating a cross-sectional configuration of a groove of a solid-state imaging device according to a modification.

(45) FIG. **32B** is a view illustrating a step of forming the groove illustrated in FIG. **32A**.

(46) FIG. **33** is a view illustrating a method for forming a concavo-convex shape of the groove.

(47) FIG. **34A** is a view illustrating a planar configuration of an etching mask according to a modification.

(48) FIG. **34B** is an enlarged view illustrating a planar configuration of a groove formed by the

etching mask illustrated in FIG. 34A.

(49) FIG. 35A is a view illustrating a planar configuration of an etching mask according to a modification.

(50) FIG. 35B is an enlarged view illustrating a planar configuration of a groove formed by the etching mask illustrated in FIG. 35A.

(51) FIG. 36 is a view illustrating a photomask for forming a resist mask illustrated in FIG. 34A.

(52) FIG. 37A is a view illustrating a photomask according to a modification.

(53) FIG. 37B is a view illustrating an etching mask formed by the photomask in FIG. 37A.

(54) FIG. 37C is an enlarged view illustrating a planar configuration of a groove formed by the etching mask illustrated in FIG. 37B.

(55) FIG. 38A is a view illustrating a photomask according to a modification.

(56) FIG. 38B is a view illustrating an etching mask formed by the photomask in FIG. 38A.

(57) FIG. 38C is an enlarged view illustrating a planar configuration of a groove formed by the etching mask illustrated in FIG. 38B.

(58) FIG. 39A is a view illustrating a photomask according to a modification.

(59) FIG. 39B is a view illustrating an etching mask formed by the photomask in FIG. 39A.

(60) FIG. 39C is an enlarged view illustrating a planar configuration of a groove formed by the etching mask illustrated in FIG. 39B.

(61) FIG. 40 is a diagram illustrating an overall configuration of an electronic device according to a third embodiment.

MODE FOR CARRYING OUT THE INVENTION

(62) Hereinafter, an example of a light detection device and an electronic device according to an embodiment of the present disclosure will be described with reference to FIGS. 1, 2, 3, 4A, 4B, 5A, 5B, 6A, 6B, 7, 8A, 8B, 9, 10, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 15, 16, 17, 18, 19, 20, 21A, 21B, 22A, 22B, 23, 24, 25, 26A, 26B, 27, 28, 29, 30, 31A, 31B, 32A, 32B, 33, 34A, 34B, 35A, 35B, 36, 37A, 37B, 37C, 38A, 38B, 38C, 39A, 39B, 39C, and 40. Embodiments of the present disclosure will be described in the following order. Note that the present disclosure is not limited to the following examples. Furthermore, the effects described in this specification are merely examples and are not limited, and other effects may be present. 1. First embodiment: solid-state imaging device 1-1 Overall configuration of solid-state imaging device 1-2 Configuration of main part 1-3 Method for manufacturing solid-state imaging device 1-4 Modification 2. Second embodiment: solid-state imaging device 2-1 Configuration of main part 2-2 Method for manufacturing solid-state imaging device 2-3 Modification 3. Third Embodiment: application to electronic device

1. First Embodiment: Solid-State Imaging Device

1-1 Overall Configuration of Solid-State Imaging Device

(63) A solid-state imaging device 1 (in a broad sense, a “light detection device”) according to a first embodiment of the present disclosure will be described. FIG. 1 is a schematic configuration diagram illustrating the entire solid-state imaging device 1 according to the first embodiment.

(64) The solid-state imaging device 1 in FIG. 1 is a back-illuminated complementary metal oxide semiconductor (CMOS) image sensor. As illustrated in FIG. 40, the solid-state imaging device 1 (101) captures image light (incident light 106) from a subject via an optical lens 102, converts a light amount of the incident light 106 formed on an imaging surface into an electrical signal on a pixel basis, and outputs as a pixel signal.

(65) As illustrated in FIG. 1, the solid-state imaging device 1 includes a substrate 2, a pixel region 3, a vertical drive circuit 4, a column signal processing circuit 5, a horizontal drive circuit 6, an output circuit 7, and a control circuit 8.

(66) The pixel region 3 has a plurality of pixels 9 arranged in a two-dimensional matrix shape on the substrate 2. The pixel 9 includes a photoelectric conversion unit 23 illustrated in FIG. 2 and a plurality of pixel transistors (not illustrated). Examples of the pixel transistor include four

transistors of a transfer transistor, a reset transistor, a selection transistor, and an amplification transistor.

(67) The vertical drive circuit **4** includes, for example, a shift register, selects a desired pixel drive wiring **10**, supplies a pulse for driving the pixels **9** to the selected pixel drive wiring **10**, and drives each pixel **9** on a row basis. That is, the vertical drive circuit **4** selectively scans each pixel **9** in the pixel region **3** sequentially in a vertical direction on a row basis, and supplies a pixel signal based on a signal charge generated in accordance with an amount of received light in the photoelectric conversion unit **23** of each pixel **9**, to the column signal processing circuit **5** through a vertical signal line **11**.

(68) The column signal processing circuit **5** is disposed, for example, for each column of the pixels **9**, and performs signal processing such as noise removal on signals outputted from the pixels **9** of one line for each pixel column. For example, the column signal processing circuit **5** performs signal processing such as correlated double sampling (CDS) for removing fixed pattern noise unique to pixels, and analog-digital (AD) conversion.

(69) The horizontal drive circuit **6** includes, for example, a shift register, sequentially outputs a horizontal scanning pulse to the column signal processing circuit **5**, sequentially selects each of the column signal processing circuits **5**, and causes each of the column signal processing circuits **5** to output a pixel signal subjected to signal processing to a horizontal signal line **12**.

(70) The output circuit **7** performs signal processing on pixel signals sequentially supplied from each of the column signal processing circuits **5** through the horizontal signal line **12**, and outputs the pixel signals. As the signal processing, for example, buffering, black level adjustment, column variation correction, various types of digital signal processing, and the like can be used.

(71) The control circuit **8** generates a clock signal and a control signal serving as a reference of operations of the vertical drive circuit **4**, the column signal processing circuit **5**, the horizontal drive circuit **6**, and the like, on the basis of a vertical synchronization signal, a horizontal synchronization signal, and a master clock signal. Then, the control circuit **8** outputs the generated clock signal and control signal to the vertical drive circuit **4**, the column signal processing circuit **5**, the horizontal drive circuit **6**, and the like.

1-2 Configuration of Main Part

(72) Next, a detailed structure of the solid-state imaging device **1** in FIG. **1** will be described. FIG. **2** is a view illustrating a cross-sectional configuration of the solid-state imaging device **1** (the pixel region **3**) in a case of being cut along line A-A in FIG. **1**.

(73) As illustrated in FIG. **2**, the solid-state imaging device **1** includes a light receiving layer **17** formed by layering the substrate **2**, a fixed charge film **13**, an insulating film **14**, a light shielding film **15**, and a planarization film **16** in this order. Furthermore, a light condensing layer **20** in which a color filter layer **18** and a microlens array **19** are layered in this order is formed on a surface of the light receiving layer **17** on the planarization film **16** side (hereinafter, also referred to as a “back surface S1 side”). Moreover, a wiring layer **21** and a support substrate **22** are layered in this order on a surface of the light receiving layer **17** on the substrate **2** side (hereinafter, also referred to as a “front surface S2 side”). Note that, since the back surface S1 of the light receiving layer **17** and a back surface of the planarization film **16** are the same surface, the back surface of the planarization film **16** is also referred to as the “back surface S1” in the following description. Furthermore, since the front surface S2 of the light receiving layer **17** and a front surface of the substrate **2** are the same surface, the front surface of the substrate **2** is also referred to as the “front surface S2” in the following description.

(74) The substrate **2** includes, for example, a semiconductor substrate made by silicon (S1), and is formed with the pixel region **3**. In the pixel region **3**, a plurality of pixels **9** is disposed in a two-dimensional matrix shape. Each pixel **9** includes the photoelectric conversion unit **23** and a plurality of pixel transistors (not illustrated). The photoelectric conversion unit **23** includes a p-type semiconductor region and an n-type semiconductor region, and a pn junction between the p-type

semiconductor region and the n-type semiconductor region constitutes a photodiode. As a result, each of the photoelectric conversion units **23** generates a signal charge corresponding to an amount of incident light on the photoelectric conversion unit **23**, and accumulates the generated signal charge in the n-type semiconductor region (a charge accumulation region).

(75) Furthermore, a pixel separation part **24** is formed between adjacent photoelectric conversion units **23**. As illustrated in FIG. **3**, the pixel separation part **24** is formed in a lattice shape on the substrate **2** so as to surround the individual pixels **9** (the photoelectric conversion units **23**). The pixel separation part **24** has a trench part **25** penetrating in a thickness direction of the substrate **2**. A side wall surface of the trench part **25** forms an outer shape of the pixel separation part **24**. That is, the trench part **25** is formed in a lattice shape on the substrate **2** so as to surround the individual photoelectric conversion units **23**. The lattice-shaped trench part **25** includes a plurality of linear trench parts **25a** extending in a row direction of the pixels **9** and a plurality of linear trench parts **25b** extending in a column direction of the pixels **9**. As illustrated in FIG. **4A**, each of the linear trench parts **25a** and **25b** is formed such that a plurality of vertical holes **26** extending in the thickness direction of the substrate **2** is aligned in a longitudinal direction of each of the linear trench parts **25a** and **25b** and adjacent vertical holes **26** are connected to each other.

(76) FIG. **4A** is an enlarged view illustrating an intersection part **27** (a portion where the linear trench part **25a** and the linear trench part **25b** intersect) of the trench part **25** of FIG. **3**. In FIG. **4A**, the fixed charge film **13** and the insulating film **14** are omitted for the intersection part **27**. Due to the plurality of vertical holes **26**, a cross-sectional shape of a side wall surface of the trench part **25** in a cross section parallel to a light incident surface (a back surface S3) of the substrate **2** is made a concavo-convex shape in which concavity and convexity are repeated along the longitudinal direction (a direction in which the trench part **25** extends) of each of the linear trench parts **25a** and **25b** constituting the trench part **25**. Furthermore, a corner part **23a** of the photoelectric conversion unit **23** at the intersection part **27** protrudes into a space in the intersection part **27**.

(77) By forming the concavo-convex shape, for example, as compared with a method of using the trench part **25** having a same trench width as a maximum width of the trench part **25** of the first embodiment and having a flat surface without concavity and convexity as a side wall surface, an area on a light incident surface side of the photoelectric conversion unit **23** can be increased by an amount of a convex part of the concavo-convex surface, and sensitivity can be increased. Furthermore, transmission of incident light can be blocked by the trench part **25**, and optical color mixing can be suppressed. Furthermore, due to the concavity and convexity of the side wall surface of the trench part **25**, for example, a reflection direction of incident light by the side wall surface can be made complicated as compared with a case where the side wall surface is a flat surface. Therefore, an optical path length of incident light entering the photoelectric conversion unit **23** can be extended, and sensitivity to a long-wavelength component (for example, infrared light or the like) can be improved.

(78) As the vertical hole **26**, for example, a cylindrical hole having a circular cross-sectional shape in the cross section parallel to the light incident surface (the back surface S3) of the substrate **2** can be adopted. In a case where the cylindrical hole is adopted, the cross-sectional shape (the concavo-convex shape) of the side wall surface of the trench part **25** is to be a wave shape in which arcs are continuous. A radius of the arc is preferably, for example, 50 nm or more and 150 nm or less. By forming the wave shape in which arcs having a radius of 50 nm or more and 150 nm or less are continuous, a reflection direction of incident light by the side wall surface of the trench part **25** can be made complicated, an optical path length of incident light can be further extended, and sensitivity to a long-wavelength component can be improved. In a case where metal is embedded in the trench part **25**, the radius of the arc is preferably made 250 nm or more.

(79) Furthermore, due to the plurality of vertical holes **26**, a plurality of concave parts **28** is formed on a bottom surface of the trench part **25** so as to be aligned in a longitudinal direction of each of the linear trench parts **25a** and **25b** constituting the trench part **25**, as illustrated in FIG. **4B**. Due to

the plurality of concave parts **28**, a cross-sectional shape of the bottom surface of the trench part **25** in a cross section perpendicular to the light incident surface (the back surface S3) of the substrate **2** is made a concavo-convex shape in which concavity and convexity are repeated along the longitudinal direction of each of the linear trench parts **25a** and **25b** constituting the trench part **25**. As the concave part **28**, for example, a parabolic concave part with a concave center can be adopted.

(80) Furthermore, the fixed charge film **13** and the insulating film **14** that cover the back surface S3 side of the substrate **2** are embedded inside the trench part **25**. Furthermore, a metal film that reflects light may be embedded in the insulating film **14**. As the metal film, for example, tungsten (W) or aluminum (Al) can be adopted.

(81) By adopting the pixel separation part **24**, each photoelectric conversion unit **23** can be shielded from light, and optical color mixing can be suppressed.

(82) The fixed charge film **13** continuously covers the entire back surface S3 side (the entire light incident surface side) of the substrate **2** and the inside of the trench part **25**. As a material of the fixed charge film **13**, for example, hafnium (Hf), aluminum (Al), zirconium (Zr), tantalum (Ta), or titanium (Ti) can be adopted. Furthermore, the insulating film **14** continuously covers an entire back surface S4 side (the entire light incident surface side) of the fixed charge film **13** and the inside of the trench part **25**. As a material of the insulating film **14**, for example, silicon oxide (SiO₂), silicon nitride (Si₃N₄), or silicon oxynitride (SiON) can be adopted.

(83) The light shielding film **15** is formed in a lattice shape that opens the light incident surface side of each of the plurality of photoelectric conversion units **23**, in a part of the insulating film **14** on a back surface S5 side so that light does not leak into the adjacent pixels **9**. Furthermore, the planarization film **16** continuously covers the entire back surface S5 side (the entire light incident surface side) of the insulating film **14** including the light shielding film **15** such that the back surface S1 of the light receiving layer **17** is a flat surface without concavity and convexity.

(84) The color filter layer **18** has a color filter **29** for each pixel **9** on the back surface S1 side (the light incident surface side) of the planarization film **16**. Each of the color filters **29** is configured to transmit wavelengths of red light, green light, blue light, and the like, and cause the transmitted incident light to be incident on the photoelectric conversion unit **23**.

(85) The microlens array **19** includes a microlens **30** for each pixel **9** on a back surface S6 side (the light incident surface side) of the color filter layer **18**. Each of the microlenses **30** is configured to condense image light (incident light) from a subject into the photoelectric conversion unit **23**.

(86) The wiring layer **21** is formed on the front surface S2 side of the substrate **2**, and includes an interlayer insulating film **31** and a wiring **32** that is layered in a plurality of layers with the interlayer insulating film **31** interposed in between. Then, the wiring layer **21** drives a pixel transistor constituting each pixel **9** via the plurality of layers of the wiring **32**.

(87) The support substrate **22** is formed on a surface of the wiring layer **21** on a side opposite to a side facing the substrate **2**. The support substrate **22** is a substrate for securing strength of the substrate **2** at a manufacturing stage of the solid-state imaging device **1**. As a material of the support substrate **22**, for example, silicon (S1) can be used.

(88) In the solid-state imaging device **1** having the above configuration, light is emitted from the back surface S1 side of the substrate **2** (the back surface S1 side of the light receiving layer **17**), the emitted light passes through the microlens **30** and the color filter **29**, and the transmitted light is photoelectrically converted by the photoelectric conversion unit **23** to generate a signal charge. Then, the generated signal charge is outputted as a pixel signal by the vertical signal line **11** illustrated in FIG. **1**, which is formed by the wiring **32**, via the pixel transistor or the like formed on the front surface S2 side of the substrate **2**.

(89) Furthermore, the solid-state imaging device **1** according to the first embodiment has been made to have a back-illuminated structure, that is, a structure in which incident light is incident from the back surface S3 side of the substrate **2** with, as a light incident surface, the back surface

S3 of the substrate 2 opposite to the front surface S2 of the substrate 2 in which the wiring layer 21 is formed. Therefore, incident light is incident on the photoelectric conversion unit 23 without being restricted by the wiring layer 21. Therefore, the opening of the photoelectric conversion unit 23 can be made wide, and for example, higher sensitivity can be achieved than that of a front-illuminated type.

1-3 Method for Manufacturing Solid-State Imaging Device

(90) Next, a method for manufacturing the solid-state imaging device 1 according to the first embodiment will be described. Here, a case of manufacturing the solid-state imaging device 1 in which a width of the trench part 25 is 160 nm will be exemplified.

(91) First, as illustrated in FIG. 5A, ion implantation of impurities is performed from the front surface S2 side (another surface side) of the substrate 2 to form the photoelectric conversion unit 23, a pixel transistor (not illustrated), and the like in the substrate 2. Subsequently, the interlayer insulating film 31 and the wiring 32 are alternately formed on the front surface S2 of the substrate 2 to form the wiring layer 21. Subsequently, the support substrate 22 is bonded to an outermost surface S7 of the wiring layer 21, and a layered body including the substrate 2, the wiring layer 21, and the support substrate 22 is formed. Subsequently, the formed layered body is inverted, and the inverted substrate 2 is polished from the back surface S3 side (one surface side) to be thinned to a desired thickness.

(92) Subsequently, as illustrated in FIG. 5B, at a boundary of the individual pixels 9 of the substrate 2, the substrate 2 is selectively etched from the back surface S3 side of the substrate 2, thereby forming the lattice-shaped trench part 25 surrounding the individual photoelectric conversion units 23. Specifically, first, as illustrated in FIGS. 6A and 6B, a resist film 33 is formed on the back surface S3 of the substrate 2, and a pattern is formed on the formed resist film 33 by a photolithography method. FIG. 6B is a view illustrating a cross section of the substrate 2 cut along line D-D in FIG. 6A. In the pattern formation, the plurality of openings 34 is formed at intervals from one another along a position where the trench part 25 is formed on the resist film 33. That is, when viewed from the back surface S3 side of the substrate 2, the plurality of openings 34 arranged in a lattice shape at predetermined intervals is formed on the resist film 33 so as to surround the individual photoelectric conversion units 23. In FIG. 6A, a case where a shape of the opening 34 is made circular is exemplified. In a case where the opening 34 of the resist film 33 is circular, the resist film 33 is exposed using a photomask 37 having a polygonal pattern 37a, as illustrated in FIG. 7. In FIG. 7, a case where a square pattern is used as the polygonal pattern 37a is exemplified. In a case where exposure is performed using the photomask 37 having the polygonal pattern 37a (for example, a square pattern), a circular pattern in which corner parts of the polygonal pattern 37a (the square pattern) are rounded is transferred to the resist film 33. Furthermore, in FIG. 6A, the opening 34 is formed at a position corresponding to a center of the intersection part 27 of the trench part 25. A diameter of the opening 34 (a circle) is 100 nm, and a distance between the openings 34 is 100 nm.

(93) Subsequently, as illustrated in FIG. 8A, anisotropic etching is performed on the substrate 2 from the back surface S3 side of the substrate 2, by using, as an etching mask, the resist film 33 in which the plurality of openings 34 is formed. As the anisotropic etching, for example, anisotropic etching using a reactive ion etching (RIE) device can be adopted. By the anisotropic etching, an opening 35 (a cylindrical opening 35) having the same cross-sectional shape as the shape of the opening 34 of the etching mask (the resist film 33) is formed on the substrate 2. A depth of the opening 35 is shallower by about 50 nm than a depth reaching an interface between the substrate 2 and the wiring layer 21. That is, the depth is reduced by a half of a distance between the openings 35. As a result, as illustrated in FIG. 8B, a plurality of openings 35 arranged in a lattice shape at predetermined intervals is formed on the substrate 2 so as to surround the individual photoelectric conversion units 23. FIG. 8B is a view illustrating a cross section of the substrate 2 cut along line E-E in FIG. 8A.

(94) Subsequently, as illustrated in FIG. 9, isotropic etching is performed on the substrate 2 from the back surface S3 side of the substrate 2, by using, as an etching mask, the etching mask (the resist film 33) used for the anisotropic etching as it is. As an etching agent for isotropic etching, for example, hydrofluoric acid can be adopted. By the isotropic etching, a side wall surface and a bottom surface of the opening 35 are etched in the substrate 2. An etching amount of the side wall surface of the opening 35 and an etching amount of the bottom surface of the opening 35 are 50 nm+over-etching of 20%. As a result, as illustrated in FIGS. 4A and 4B, adjacent openings 35 (vertical holes 26) are connected to each other, so that the lattice-shaped trench part 25 is formed on the substrate 2 so as to surround the photoelectric conversion units 23. The trench part 25 is formed such that the plurality of openings 35 (vertical holes 26) extending in the thickness direction of the substrate 2 is aligned in the longitudinal direction of each of the linear trench parts 25a and 25b constituting the trench part 25, and adjacent vertical holes 26 are connected to each other. Therefore, for example, as compared with a method of using the trench part 25 having the same trench width as the maximum width of the trench part 25 of the first embodiment and having a flat surface without concavity and convexity as a side wall surface, an area on the light incident surface side of the photoelectric conversion unit 23 can be increased by an amount of the convex part of the concavo-convex surface, and the sensitivity can be further increased. Furthermore, a reflection direction of incident light by the side wall surface of the trench part 25 can be made complicated, an optical path length of incident light entering the photoelectric conversion unit 23 can be extended, and sensitivity to a long-wavelength component (for example, infrared light or the like) can be improved. Note that, since an over-etching amount of the bottom surface of the trench part 25 is 20% of 50 nm, it is about 10 nm.

(95) Subsequently, the etching mask is removed from the substrate 2, the fixed charge film 13 and the insulating film 14 are formed so as to continuously cover the side wall surface and the bottom surface of the trench part 25 and the entire back surface S3 side (the light receiving surface side) of the substrate 2, and the pixel separation part 24 having the trench part 25 is formed. Subsequently, after a light-shielding material layer is formed on the entire back surface S5 side of the insulating film 14, the light-shielding material layer is patterned into a desired shape. Thereafter, by forming the planarization film 16, the color filter layer 18, and the microlens array 19 on the back surface S3 side of the substrate 2, the solid-state imaging device 1 illustrated in FIG. 2 is completed.

(96) As described above, in the method for manufacturing the solid-state imaging device 1 according to the first embodiment, first, by performing anisotropic etching from the back surface S3 side (one surface side) of the substrate 2, the plurality of openings 35 arranged at predetermined intervals from one another is formed on the substrate 2 so as to surround each photoelectric conversion unit 23. Subsequently, by performing isotropic etching from the back surface S3 side (one surface side) of the substrate 2 to connect the adjacent openings 35, the trench part 25 is formed along an arrangement direction of the openings 35 on the substrate 2 so as to surround the individual photoelectric conversion units 23.

(97) In other words, it can be said that the plurality of openings 35 arranged at predetermined intervals from one another is formed on the substrate 2 by performing anisotropic etching from one surface side of the substrate 2, and then the trench part 25 is formed along the arrangement direction of the openings 35 on the substrate 2 by performing isotropic etching from one surface side of the substrate 2 to connect the adjacent openings 35 to each other.

(98) As a result, the corner part 23a of the intersection part 27 of the trench part 25 can be suppressed from being rounded, and an area of the corner part 23a can be increased. Therefore, vignetting of incident light by the intersection part 27 can be suppressed, and sensitivity can be improved. Furthermore, transmission of incident light can be blocked by the trench part 25, and optical color mixing can be suppressed. Therefore, it is possible to manufacture the solid-state imaging device 1 capable of suppressing optical color mixing while improving sensitivity. That is, it is possible to manufacture the solid-state imaging device 1 capable of improving characteristics

such as improvement of sensitivity and reduction of optical color mixing.

(99) Furthermore, according to Rayleigh's equation: resolution (minimum pattern)= $k_1 \cdot \lambda / NA$, a resolution of a transfer image by a light exposure device is to be resolution (minimum pattern)=35.7 nm, in a case where $k_1=0.25$, $\lambda=193$, and $NA=1.35$ are assumed. However, in practice, 35.7 nm is difficult to achieve by the current process technology, and a current guaranteed minimum line width is 55 nm for a line and 90 nm for a hole in a case where an ArF laser is used, and 35 nm for a line and 62.5 nm for a hole in a case where immersion is used. Therefore, in the method for manufacturing the solid-state imaging device **1** according to the first embodiment, in a case where the width of the trench part **25** is minimized, as illustrated in FIG. **10**, an etching mask (the resist film **33**) is used in which the opening **34** is a hole (a circle) having a diameter of 62.5 nm and a distance between the openings **34** is 35 nm. In this case, when the openings **35** come into contact with each other by the isotropic etching, the diameter of the opening **35** is to be 97.5 nm. Furthermore, when the over-etching of 20% is finished, the diameter of the opening **35** is to be about 105 nm. In FIG. **10**, the opening **35** after the over-etching of 20% is finished is indicated by a broken line. Here, for example, according to a conventional method of forming the trench part **25** by using an etching mask having lattice-shaped openings, the minimum line width of the trench part **25** is 100 nm to 110 nm. Therefore, according to the solid-state imaging device **1** according to the first embodiment, it is possible to increase an area of the photoelectric conversion unit **23** on the light incident surface side while achieving a width similar to the minimum line width by the conventional method, and it is possible to improve the sensitivity.

(100) Furthermore, in a case where a metal such as tungsten (W) is embedded in the trench part **25**, a width of 500 nm or more is required, but the sensitivity can be similarly improved even in a case where the width is 500 nm or more.

(101) Furthermore, since the trench part **25** is formed by connecting the openings **35** after the openings **35** are formed, a microloading effect can be prevented, excessive etching of the intersection part **27** can be prevented, and the depth of the trench part **25** can be made uniform. Therefore, for example, in a case where the substrate **2** is etched by an amount of a thickness of the substrate **2**, the excessive depth of the intersection part **27** can be prevented.

(102) In this regard, for example, according to a method of forming the trench part **25** by using the etching mask having lattice-shaped openings, etching in a depth direction of the intersection parts **27** becomes excessive due to the microloading effect. Specifically, for example, in a case where over-etching of 5 nm is performed on portions other than the intersection part **27**, over-etching of 15 nm is performed on the intersection part **27**. Therefore, for example, if an etching time is reduced in consideration of the etching amount of the intersection part **27**, a portion other than the intersection part **27** is not etched to a sufficient depth, and a structure disadvantageous for suppressing optical color mixing and the like is obtained.

(103) Furthermore, in the method for manufacturing the solid-state imaging device **1** according to the first embodiment, anisotropic etching and isotropic etching have been performed by using the etching mask formed on the back surface S3 (one surface) of the substrate **2** and having the plurality of openings **34** arranged in a lattice shape at predetermined intervals so as to surround the individual photoelectric conversion units **23** when viewed from the one surface side of the substrate **2**. As a result, a photolithography step and an etching step (a step including both anisotropic etching and isotropic etching) only need to be performed once, so that an increase in the number of steps can be suppressed. Furthermore, rectangularity of the corner part **23a** of the intersection part **27** of the trench part **25** and a shape of the side wall surface of the trench part **25** can be easily changed by a shape (a pattern shape) of the opening **34** of the etching mask.

(104) In this regard, for example, in a method of forming the linear trench parts **25a** with anisotropic etching by using an etching mask having only openings corresponding to the linear trench parts **25a** extending in the row direction of the pixels **9**, and then forming the linear trench parts **25b** with anisotropic etching by using an etching mask having only openings corresponding to

the linear trench parts **25b** extending in the column direction, the corner part **23a** of the intersection part **27** of the trench part **25** can be suppressed from being rounded, but it is necessary to perform each of the photolithography step and the etching step twice, and the number of steps increases. (105) Furthermore, in the solid-state imaging device **1** according to the first embodiment, the cross-sectional shape of the side wall surface of the trench part **25** in the cross section parallel to the back surface S3 (the light incident surface) of the substrate **2** has been made a concavo-convex shape in which concavity and convexity are repeated along the longitudinal direction of each of the linear trench parts **25a** and **25b** constituting the trench part **25**. As a result, for example, as compared with a method of using the trench part **25** having the same trench width as the maximum width of the trench part **25** of the first embodiment and having a flat surface without concavity and convexity as a side wall surface, an area on the light incident surface side of the photoelectric conversion unit **23** can be increased by an amount of the convex part of the concavo-convex surface, and the sensitivity can be increased. Furthermore, transmission of incident light can be blocked by the trench part **25**, and optical color mixing can be suppressed. Therefore, it is possible to provide the solid-state imaging device **1** capable of suppressing optical color mixing while improving sensitivity. That is, it is possible to provide the solid-state imaging device **1** capable of improving characteristics such as improvement of sensitivity and reduction of optical color mixing.

1-4 Modification

(106) (1) Note that, in the first embodiment, an example has been described in which the etching mask (the resist film **33**) is removed after isotropic etching, but other configurations can also be adopted. For example, a configuration may also be adopted in which isotropic etching and removal of the etching mask (the resist film **33**) are performed simultaneously. That is, the step of removing the resist film **33** may be immediately started after the anisotropic etching is finished, and the isotropic etching of the substrate **2** may be performed together with the removal of the resist film **33**.

(107) Furthermore, an example has been described in which a resist mask made with the resist film **33** is used as the etching mask, but other configurations can also be adopted. For example, a hard mask may be used.

(108) (2) Furthermore, in the first embodiment, an example has been described in which the cross-sectional shape of the side wall surface of the trench part **25** in the cross section parallel to the light incident surface (the back surface S3) of the substrate **2** is made a concavo-convex shape (hereinafter, also referred to as a “first concavo-convex shape”) in which concavity and convexity are repeated along the longitudinal direction of each of the linear trench parts **25a** and **25b** constituting the trench part **25**, but other configurations may also be adopted. For example, as illustrated in FIG. **11A**, in addition to the first concavo-convex shape, the cross-sectional shape of the side wall surface of the trench part **25** in the cross section perpendicular to the light incident surface (the back surface S3) of the substrate **2** may be made a concavo-convex shape (hereinafter, also referred to as a “second concavo-convex shape”) in which concavity and convexity are repeated along a depth direction of the trench part **25**. With the configuration having the second concavo-convex shape, a reflection direction of incident light by the side wall surface of the trench part **25** can be made further complicated, an optical path length of incident light can be further extended, and the sensitivity to a long-wavelength component can be improved.

(109) As a method of forming the second concavo-convex shape, for example, a method can be adopted in which a concave part called a scallop is formed on the side wall surface of the opening **35** as illustrated in FIG. **11B**, by performing anisotropic etching performed in the step of forming the trench part **25** by using a Bosch process. The Bosch process is a method of forming a deep and narrow groove on the substrate **2** by repeating etching of the substrate **2**, formation of a protective film in a concave part formed by the etching, and removal of the protective film from a bottom surface of the concave part. Since the etching selectively proceeds in a depth direction of the substrate **2**, the etching by the Bosch process can be said to be substantially anisotropic etching. By

forming the scallop on the side wall surface of the opening 35, as illustrated in FIG. 11A, the opening 35 is widened while concavity and convexity of the scallop are maintained in isotropic etching, and the second concavo-convex shape can be formed.

(110) (3) Furthermore, in the first embodiment, an example has been described in which the shape of the opening 34 of the etching mask (the resist film 33) is made circular, but other configurations can also be adopted. For example, the shape may be an elliptical shape as illustrated in FIG. 12A, or may be a polygonal shape such as a rectangular shape or a hexagonal shape as illustrated in FIGS. 13A and 14A. In FIGS. 12A, 13A, and 14A, a case is exemplified in which the elliptical or polygonal openings 34 extend along an arrangement direction of the openings 34. Furthermore, in FIGS. 12A, 12B, 13A, 13B, and 14A, the opening 34 is not formed at a position corresponding to a center of the intersection part 27.

(111) Note that, in a case where the opening 34 of the resist film 33 has an elliptical shape, as illustrated in FIG. 15, the resist film 33 is exposed by using the photomask 37 having the polygonal pattern 37a in which a square and a rectangle are superimposed on each other. In a case where exposure is performed using the photomask 37 having the polygonal pattern 37a illustrated in FIG. 15, as illustrated in FIG. 12A, an elliptical pattern in which corner parts of polygonal pattern 37a are rounded is transferred to the resist film 33. Furthermore, in a case where the opening 34 of the resist film 33 has a polygonal shape such as a rectangular or a hexagon, the resist film 33 is exposed using the photomask 37 having the polygonal pattern 37a having the same shape as the opening 34. Here, a guaranteed minimum line width of the opening 34 that can be formed on the resist film 33 is a line of 35 nm and a hole of 62.5 nm in a case where immersion is used. Therefore, as illustrated in FIGS. 12B, 13B, and 14B, for example, when an elliptical shape, a rectangular shape, or a hexagonal shape is used as the shape of the opening 34 of the resist film 33, a maximum width of the trench part 25 can be narrowed as compared with a case of using a circular shape.

(112) (4) Furthermore, in the first embodiment, an example has been described in which the microlens 30 is formed for each photoelectric conversion unit 23, but other configurations can also be adopted. For example, as illustrated in FIG. 16, a configuration may also be adopted in which the microlens 30 is formed for each photoelectric conversion unit group 36 including the photoelectric conversion units 23 disposed in a 2×2 matrix shape when viewed from the back surface S3 side of the substrate 2. Here, in the case of the configuration in which the microlens 30 is formed for each photoelectric conversion unit group 36, since incident light is condensed on the intersection part 27 of the trench part 25, vignetting of incident light easily occurs. On the other hand, in the solid-state imaging device 1 according to the first embodiment, since the corner part 23a of the intersection part 27 of the trench part 25 is suppressed from being rounded, an area of the corner part 23a can be increased, and vignetting of incident light can be suppressed. Note that the configuration in which the microlens 30 is formed for each photoelectric conversion unit group 36 may be applied to all the photoelectric conversion units 23 in the pixel region 3, or may be applied only to some of the photoelectric conversion units 23.

(113) (5) Furthermore, in the first embodiment, an example has been described in which the solid-state imaging device 1 (a CMOS image sensor) is used as the light detection device, but other configurations can also be adopted. For example, as illustrated in FIG. 17, a distance measuring sensor may be used in which the microlens array 19 is layered on the back surface S3 side (the light incident surface side) of the substrate 2 without interposing the color filter layer 18 (the plurality of color filters 29). In the distance measuring sensor, since the color filter 29 is omitted unlike the CMOS image sensor, infrared light is incident on the photoelectric conversion unit 23 as incident light. As the distance measuring sensor, for example, a distance image sensor such as a single photon avalanche diode (SPAD) sensor used for distance measurement by a time of flight (ToF) method can be adopted. Here, in a case where the light detection device is a distance measuring sensor, incident light (infrared light) is less likely to be absorbed by the photoelectric conversion

unit **23** as compared with a case where the light detection device is the solid-state imaging device **1** (the CMOS image sensor). On the other hand, in the first embodiment, since an optical path length of incident light (infrared light) is extended by the photoelectric conversion unit **23**, incident light (infrared light) is easily absorbed by the photoelectric conversion unit **23**, and decrease in sensitivity can be suppressed.

(114) Note that, also in a case where the light detection device is a distance measuring sensor, as illustrated in FIG. **18**, the trench part **25** can be formed on the substrate **2** by performing the anisotropic etching to form the openings **35** on the substrate **2** and then performing the isotropic etching to connect the openings **35**, similarly to a case where the light detection device is a CMOS image sensor. Furthermore, as illustrated in FIG. **19**, by forming the cross-sectional shape of the side wall surface of the trench part **25** in the cross section perpendicular to the light incident surface (the back surface S3) of the substrate **2** into a concavo-convex shape in which concavity and convexity are repeated along the depth direction of the trench part **25**, a reflection direction of incident light by the side wall surface of the trench part **25** can be made further complicated, and an optical path length of incident light can be further extended.

(115) <2. Second embodiment: solid-state imaging device>[2-1 Configuration of main part] Next, a solid-state imaging device according to a second embodiment of the present disclosure will be described. An overall configuration of the solid-state imaging device according to the second embodiment is similar to that in FIG. **1**, and thus illustration thereof is omitted. FIG. **20** is a view illustrating an overall configuration of a chip **38** in which a solid-state imaging device **1** is formed. FIG. **21A** is an enlarged view of a region F in FIG. **20**. FIG. **21B** is a view illustrating a cross-sectional configuration of the solid-state imaging device **1** (the chip **38**) in a case of being cut along line G-G in FIG. **21A**. In FIGS. **20**, **21A**, and **21B**, portions corresponding to FIGS. **1**, **2**, **3**, **4A**, **4B**, **5A**, **5B**, **6A**, **6B**, **7**, **8A**, **8B**, **9**, **10**, **11A**, **11B**, **12A**, **12B**, **13A**, **13B**, **14A**, **14B**, **15**, **16**, **17**, **18**, and **19** are denoted by the same reference numerals, and redundant description is omitted.

(116) As illustrated in FIG. **20**, the solid-state imaging device **1** according to the second embodiment is different from that of the first embodiment in that a groove **39** is formed so as to surround a pixel region **3**. Furthermore, as illustrated in FIG. **21B**, the chip **38** formed with the solid-state imaging device **1** includes a sensor substrate **40** and a logic substrate **41** bonded to the sensor substrate **40**. The sensor substrate **40** includes a light receiving layer **17**, a light condensing layer **20**, and a wiring layer **21**, and a configuration of a trench part **25** is the same as that of the first embodiment (the configuration illustrated in FIGS. **1**, **2**, **3**, **4A**, **4B**, **5A**, **5B**, **6A**, **6B**, **7**, **8A**, **8B**, **9**, **10**, **11A**, **11B**, **12A**, **12B**, **13A**, **13B**, **14A**, **14B**, **15**, **16**, **17**, **18**, and **19**). Furthermore, the logic substrate **41** includes a vertical drive circuit **4**, a column signal processing circuit **5**, a horizontal drive circuit **6**, an output circuit **7**, and a control circuit **8** illustrated in FIG. **1**. The chip **38** is obtained by forming a wafer in which a plurality of chips **38** is arranged in a two-dimensional matrix shape and then dicing (dividing) the wafer with a blade.

(117) Furthermore, as illustrated in FIG. **20**, on an outer peripheral side of the chip **38**, a blade region **42** after dividing is formed so as to surround the pixel region **3**. The blade region **42** after dividing is a region formed between the individual chips **38** in the wafer and obtained by dicing, with a blade, a bottom surface of a groove-shaped region (the blade region **42** before dividing) opened on a back surface S3 side of a substrate **2**. Furthermore, the groove **39** is formed between the blade region **42** after dividing and the pixel region **3** so as to surround the pixel region **3**. The groove **39** penetrates the substrate **2** in a thickness direction, and a surface S8 (hereinafter, also referred to as a “back surface S8”) of the wiring layer **21** facing the substrate **2** forms a bottom surface of the groove **39**.

(118) Similarly to the trench part **25**, as illustrated in FIG. **22A**, the groove **39** is formed such that a plurality of vertical holes **43** extending in the thickness direction of the substrate **2** is aligned in a direction in which the groove **39** extends (in FIG. **22A**, a vertical direction), and adjacent vertical holes **43** are connected to each other. FIG. **22A** is an enlarged view illustrating a linear portion

constituting the groove **39** in FIG. **20**. Due to the plurality of the vertical holes **43**, a cross-sectional shape of a side wall surface of the groove **39** in a cross section parallel to a light incident surface (the back surface S3) of the substrate **2** is made a concavo-convex shape in which concavity and convexity are repeated along the direction in which the groove **39** extends. A maximum width of the groove **39** is preferably 500 nm or less, more preferably 400 nm or less, and still more preferably 200 nm or less. Furthermore, a lower limit of the maximum width of the groove **39** is preferably 100 nm or more.

(119) As the vertical hole **43**, for example, a cylindrical hole having a circular cross-sectional shape in the cross section parallel to the light incident surface (the back surface S3) of the substrate **2** can be adopted. In a case where the cylindrical hole is adopted, the cross-sectional shape (the concavo-convex shape) of the side wall surface of the groove **39** is to be a wave shape in which arcs are continuous. A radius of the arc is preferably, for example, 250 nm or less. By forming the wave shape in which arcs having a radius of 250 nm or less are continuous, the maximum width of the groove **39** can be set to 500 nm or less. Furthermore, a reflection direction of light by the side wall surface of the groove **39** can be made complicated, and incident light can be scattered in a case where light enters the groove **39**.

(120) Furthermore, due to the plurality of vertical holes **43**, a plurality of concave part **44** is formed on the bottom surface of the groove **39** so as to be aligned in the direction in which the groove **39** extends (a horizontal direction in FIG. **22B**) as illustrated in FIG. **22B**. Due to the plurality of concave part **44**, a cross-sectional shape of the bottom surface of the groove **39** in a cross section perpendicular to the light incident surface (the back surface S3) of the substrate **2** is made a concavo-convex shape in which concavity and convexity are repeated along the direction in which the groove **39** extends. As the concave part **28**, for example, a parabolic concave part with a concave center can be adopted.

(121) Here, for example, in the solid-state imaging device **1** that does not include the groove **39**, when the wafer is diced by a blade, if the blade comes into contact with the substrate **2** and film peeling or a crack occurs in the substrate **2**, the film peeling or the like of the substrate **2** progresses, and the film peeling or the like may progress into the pixel region **3**.

(122) On the other hand, in the solid-state imaging device **1** according to the second embodiment, since the groove **39** is provided between the blade region **42** after dividing and the pixel region **3**, the film peeling or the like of the substrate **2** occurring at the time of dicing can be stopped by the groove **39**, and progress of the film peeling or the like into the pixel region **3** can be prevented.

(123) Furthermore, for example, in the solid-state imaging device **1** in which the groove **39** surrounding the pixel region **3** is provided but the side wall surface of the groove **39** does not have a concavo-convex shape (the side wall surface is a flat surface), when light is incident on the groove **39**, there is a possibility that the light is reflected by the bottom surface of the groove **39** and the side wall surface of the groove **39** and travels toward the pixel region **3**. Then, unnecessary light may be incident on the pixel region **3** to cause flare.

(124) On the other hand, the inventor of the present disclosure has found from daily research that such flare is suppressed by narrowing a width of the groove **39**. Then, on the basis of this finding, the inventor has found that no flare problem occurs by setting the width of the groove **39** to 500 nm or less. Note that, from the viewpoint of improving productivity, a current minimum width of the groove **39** is 1800 nm. On the basis of such a finding, in the solid-state imaging device **1** according to the second embodiment, the maximum width of the groove **39** has been set to 500 nm or less. Therefore, it is possible to suppress an occurrence of flare due to light incident into the groove **39**.

(125) Furthermore, in the solid-state imaging device **1** according to the second embodiment, the cross-sectional shape of the side wall surface of the groove **39** in the cross section parallel to the back surface S3 of the substrate **2** is made a concavo-convex shape in which concavity and convexity are repeated along the direction in which the groove **39** extends. Therefore, light incident into the groove **39** can be scattered, incident light reflected by the side wall surface of the groove

39 can be suppressed from returning to the pixel region **3** side, and an occurrence of flare can be further suppressed.

(126) Furthermore, the cross-sectional shape of the bottom surface of the groove **39** in the cross section perpendicular to the back surface S3 (the light incident surface) of the substrate **2** has been made a concavo-convex shape in which concavity and convexity are repeated along the direction in which the groove **39** extends. As a result, light incident into the groove **39** can be scattered more strongly, light reflected by the side wall surface of the groove **39** can be suppressed from returning to the pixel region **3** side, and an occurrence of flare can be further suppressed. Furthermore, as described in the first embodiment, an occurrence of flare can be further suppressed by embedding a metal film (tungsten (W) or aluminum (AL)) inside the trench part **25** between the photoelectric conversion units **23**.

2-2 Method for Manufacturing Solid-State Imaging Device

(127) Next, a method for manufacturing the solid-state imaging device **1** according to the second embodiment will be described. Here, only a step of forming the groove **39**, which is not included in the manufacturing method described in the first embodiment, will be described.

(128) The groove **39** is formed simultaneously with the trench part **25** surrounding the photoelectric conversion units **23**. That is, the groove **39** is formed in the same step as the step of forming the trench part **25**. Specifically, first, in a step of forming a pattern on a resist film **33**, on the resist film **33**, the plurality of openings **34** is formed at intervals from one another along a position where the trench part **25** is formed as illustrated in FIGS. **6A**, **6B**, and **23**, and a plurality of openings **45** is formed at intervals from one another along a position where the groove **39** is formed as illustrated in FIGS. **23** and **24**. That is, when viewed from the back surface S3 side of the substrate **2**, on the resist film **33**, the plurality of openings **34** arranged in a lattice shape at predetermined intervals is formed so as to surround the individual photoelectric conversion units **23**, and the plurality of openings **45** arranged in a lattice shape at predetermined intervals is formed so as to surround each pixel region **3**. FIG. **6A** is a view illustrating a region where the opening **34** is formed.

(129) In FIGS. **6A** and **23**, a case where shapes of the openings **34** and **45** are made circular is exemplified. In a case where the openings **34** and **45** of the resist film **33** are circular, as illustrated in FIG. **25**, the resist film **33** is exposed by using a photomask having polygonal patterns **37a** and **37b** as a photomask **37**. In FIG. **25**, a case is exemplified in which a square pattern is used as the polygonal patterns **37a** and **37b**, and the polygonal pattern **37b** (the square pattern) corresponding to the opening **45** is slightly larger than the polygonal pattern **37a** (the square pattern) corresponding to the opening **34**. In a case where exposure is performed using the photomask **37** having the polygonal patterns **37a** and **37b**, a circular pattern in which corner parts of the polygonal patterns **37a** and **37b** are rounded is transferred to the resist film **33**.

(130) Subsequently, as illustrated in FIGS. **8A** and **26A**, anisotropic etching (RIE or the like) is performed on the substrate **2** from the back surface S3 side of the substrate **2** by using, as an etching mask, the resist film **33** in which the plurality of openings **34** and **45** is formed. By the anisotropic etching, openings **35** and **46** (cylindrical openings **35** and **46**) having the same cross-sectional shape as the shapes of the openings **34** and **45** of the etching mask (the resist film **33**) are formed on the substrate **2**. As a result, as illustrated in FIG. **8B**, a plurality of openings **35** ("first openings" in a broad sense) arranged in a lattice shape at predetermined intervals is formed on the substrate **2** so as to surround the individual photoelectric conversion units **23**. At the same time, as illustrated in FIG. **26B**, a plurality of openings **46** ("second openings" in a broad sense) arranged at predetermined intervals is formed on the substrate **2** so as to surround each pixel region **3**. In FIG. **23**, a case is exemplified in which the interval between the openings **35** and the interval between the openings **46** are the same. Furthermore, FIG. **26B** is a view illustrating a cross section of the substrate **2** cut along line J-J in FIG. **26A**.

(131) Subsequently, as illustrated in FIGS. **9** and **27**, isotropic etching is performed on the substrate **2** from the back surface S3 side of the substrate **2**, by using, as an etching mask, the etching mask

(the resist film 33) used for the anisotropic etching as it is. As an etching agent for isotropic etching, for example, hydrofluoric acid can be adopted. By the isotropic etching, a side wall surface and a bottom surface of the opening 35 and a side wall surface and a bottom surface of the opening 46 are etched in the substrate 2. As a result, as illustrated in FIGS. 4A and 4B, adjacent openings 35 (vertical holes 26) are connected to each other, so that the lattice-shaped trench part 25 is formed on the substrate 2 so as to surround the photoelectric conversion units 23. Similarly, as illustrated in FIGS. 22A and 22B, the groove 39 is formed on the substrate 2 so as to surround each pixel region 3 by connecting adjacent openings 46 (vertical holes 43) to each other. In the groove 39, the plurality of openings 46 (vertical holes 43) extending in the thickness direction of the substrate 2 is aligned in the direction in which the groove 39 extends, and adjacent vertical holes 43 are connected to each other. Therefore, a reflection direction of incident light by the side wall surface of the groove 39 can be made complicated, and incident light incident on the groove 39 can be scattered.

(132) Note that the present method of etching the groove 39 simultaneously with etching of the trench part 25 can also be adopted for another light detection device such as a SPAD sensor (see FIG. 17) used for distance measurement by the ToF method, for example, as long as the light detection device is a light detection device in which the trench part 25 penetrates the substrate 2.

(133) As described above, in the method for manufacturing the solid-state imaging device 1 according to the second embodiment, first, simultaneously with formation of the plurality of openings 35 (first openings) arranged so as to surround each photoelectric conversion unit 23 on the substrate 2 by anisotropic etching, the plurality of openings 46 (second openings) arranged at predetermined intervals is formed so as to surround the pixel region 3 having the plurality of photoelectric conversion units 23. Subsequently, simultaneously with formation of the trench part 25 on the substrate 2 by connecting adjacent openings 35 to each other by isotropic etching, adjacent openings 46 are connected to each other to form the groove 39 so as to surround the pixel region 3.

(134) In other words, it can be said that the anisotropic etching is performed from one surface side of the substrate 2 to form the plurality of openings 35 arranged at predetermined intervals from one another on the substrate 2, and then isotropic etching is performed from one surface side of the substrate 2 to connect adjacent openings 35, to form the groove 39 on the substrate 2 along the arrangement direction of the openings 35.

(135) As a result, it is possible to provide the solid-state imaging device 1 in which the groove 39 can be simultaneously formed in the step of forming the trench part 25, the width of the groove 39 can be made equal to or less than a threshold value at which a flare problem does not occur while suppressing decrease in productivity, and an occurrence of flare due to light incident in the groove 39 can be suppressed. That is, it is possible to manufacture the solid-state imaging device 1 capable of improving characteristics such as image quality improvement by suppressing flare.

(136) Specifically, first, a current minimum line width is a line of 35 nm and a hole of 62.5 nm in a case where immersion is used. Therefore, in the method for manufacturing the solid-state imaging device 1 according to the second embodiment, in a case where the width of the groove 39 is minimized, as illustrated in FIG. 28, an etching mask (the resist film 33) is used in which the opening 45 is a hole (a circle) having a diameter of 62.5 nm and a distance between the openings 45 is 35 nm. In this case, when the openings 46 of the substrate 2 come into contact with each other by the isotropic etching, the diameter of the opening 46 is to be 97.5 nm. Furthermore, when over-etching of 20% is finished, the diameter of the opening 46 is to be about 105 nm. In FIG. 28, the opening 35 after the over-etching of 20% is finished is indicated by a broken line. Therefore, according to the solid-state imaging device 1 according to the second embodiment, the width of the groove 39 can be set to $\frac{1}{5}$ of a threshold value (500 nm) at which a flare problem does not occur, and an occurrence of flare can be suppressed.

(137) Here, for example, in a case where one of the side wall surfaces of the groove 39 has a saw-

tooth shape as illustrated in FIG. 29 in order to suppress an occurrence of flare, the width of the groove 39 is to be 11720 nm when a width of each tooth is 3.5 μm , a tip angle is 20°, and a distance between a tip end of the sawtooth shape and another side wall surface is 1800 nm. Therefore, for example, since a region of the groove 39 is larger than that of a case where the side wall surface of the groove 39 does not have a sawtooth shape, it is necessary to change a wiring design under the region of the groove 39.

(138) On the other hand, in the method for manufacturing the solid-state imaging device 1 according to the second embodiment, the width of the groove 39 can be narrowed, and the region of the groove 39 can be reduced. Therefore, the groove 39 can be accommodated within a margin area, and a wiring design under the region of the groove 39 does not need to be changed.

(139) Furthermore, according to the sawtooth shape illustrated in FIG. 29, an occurrence of flare can be suppressed, but the photomask 37 deviates from a mask quality assurance standard.

(140) Therefore, for example, in a case where manufacturing is requested to a mask shop (a reticle maker), there is a possibility that, even if the photomask 37 has a defect, a guarantee such as recreating of the photomask 37 may not be given by the mask shop.

(141) On the other hand, in the solid-state imaging device 1 according to the second embodiment, the photomask 37 can be designed with a pattern such as a square pattern that does not deviate from the mask quality assurance standard. Therefore, if the photomask 37 has a defect, the guarantee can be received from the mask shop.

(142) Furthermore, in the method for manufacturing the solid-state imaging device 1 according to the second embodiment, anisotropic etching and isotropic etching have been performed by using the etching mask (the resist film 33) formed on the back surface S3 (one surface) of the substrate 2 and having the plurality of openings 45 arranged at predetermined intervals so as to surround the pixel region 3 when viewed from the one surface side of the substrate 2. As a result, a photolithography step and an etching step (a step including both anisotropic etching and isotropic etching) only need to be performed once, so that an increase in the number of steps can be suppressed.

2-3 Modification

(143) (1) Note that, in the second embodiment, an example has been described in which the etching of the trench part 25 and the etching of the groove 39 are simultaneously performed, but other configurations can also be adopted. For example, in a case of manufacturing the solid-state imaging device 1 (a CMOS image sensor or the like) in which the trench part 25 does not penetrate the substrate 2, an etching depth of the substrate 2 is different between the position where the trench part 25 is formed and the position where the groove 39 is formed. Therefore, it is necessary to perform etching of the trench part 25 and etching of the groove 39 in separate steps. That is, in a step different from the step of forming the trench part 25, the groove 39 is formed so as to surround the pixel region 3 by performing anisotropic etching on the substrate 2 from one surface side of the substrate 2 to form the plurality of openings 46 (second openings) arranged at predetermined intervals on the substrate 2 so as to surround the pixel region 3, and then performing isotropic etching on the substrate 2 from one surface side of the substrate 2 to connect adjacent openings 46 to each other.

(144) For example, after the step of forming the trench part 25, the step of forming the groove 39 is performed. Specifically, first, a resist film 47 is formed on the back surface S3 of the substrate 2 after formation of the trench part 25, and the plurality of openings 45 is formed at intervals from one another along the position where the groove 39 is formed, as illustrated in FIG. 30. That is, when viewed from the back surface S3 side of the substrate 2, the plurality of openings 45 arranged in a lattice shape at predetermined intervals is formed on the resist film 47 so as to surround each pixel region 3. Subsequently, as illustrated in FIG. 31A, anisotropic etching is performed on the substrate 2 from the back surface S3 side of the substrate 2, by using, as an etching mask, the resist film 47 in which the plurality of openings 45 is formed. By the anisotropic etching, the plurality of

openings **46** arranged at predetermined intervals is formed on the substrate **2** so as to surround each pixel region **3**. Subsequently, as illustrated in FIG. **31B**, isotropic etching is performed on the substrate **2** from the back surface S3 side of the substrate **2**, by using the etching mask (the resist film **47**) used for the anisotropic etching as it is. By the isotropic etching, as illustrated in FIGS. **22A** and **22B**, the groove **39** is formed on the substrate **2** so as to surround each pixel region **3** by connecting adjacent openings **46** (vertical holes **43**) to each other.

(145) (2) Furthermore, in the second embodiment, an example has been described in which the cross-sectional shape of the side wall surface of the groove **39** in the cross section parallel to the light incident surface (the back surface S3) of the substrate **2** is made a concavo-convex shape (hereinafter also referred to as a “third concavo-convex shape”) in which concavity and convexity are repeated along the direction in which the groove **39** extends, but other configurations may also be adopted. For example, in addition to the third concavo-convex shape, as illustrated in FIG. **32A**, the cross-sectional shape of the side wall surface of the groove **39** in the cross section perpendicular to the light incident surface (the back surface S3) of the substrate **2** may be made a concavo-convex shape (hereinafter, also referred to as a “fourth concavo-convex shape”) in which concavity and convexity are repeated along a depth direction of the groove **39**. By having the fourth concavo-convex shape, a reflection direction of light by the side wall surface of the groove **39** can be made further complicated, light incident on the groove **39** can be more strongly scattered, incident light reflected by the side wall surface of the groove **39** can be suppressed from returning to the pixel region **3** side, and an occurrence of flare can be further suppressed.

(146) As a method of forming the fourth concavo-convex shape, for example, a method can be adopted in which a concave part called a scallop is formed on the side wall surface of the opening **46** as illustrated in FIG. **32B**, by performing anisotropic etching performed on the step of forming the groove **39** by using a Bosch process. By forming the scallop on the side wall surface of the opening **46**, the opening **46** is widened while concavity and convexity of the scallop are maintained in the isotropic etching, and the fourth concavo-convex shape can be formed on the side wall surface of the groove **39** as illustrated in FIG. **32A**. As a result, the formation of the groove **39** and the formation of the fourth concavo-convex shape on the side wall surface of the groove **39** can be simultaneously performed, and an increase in the number of steps can be suppressed.

(147) In this regard, for example, as illustrated in FIG. **33**, in a method of forming the fourth concavo-convex shape by isotropic etching using an etching mask **48** having an opening corresponding to a concave part of the fourth concavo-convex shape after forming the groove **39** not having the fourth concavo-convex shape, the number of steps increases.

(148) (3) Furthermore, in the second embodiment, an example has been described in which the shape of the opening **45** of the etching mask (the resist film **33**) is made circular, but other configurations can also be adopted. For example, the shape may be an elliptical shape as illustrated in FIG. **34A**, or may be a polygonal shape such as a rectangular shape or a hexagonal shape as illustrated in FIG. **35A**. In FIGS. **34A** and **35A**, a case is exemplified in which the elliptical or polygonal opening **45** extends along a width direction of the groove **39** (a direction orthogonal to the direction in which the groove **39** extends).

(149) Note that, in a case where the opening **45** of the resist film **33** has an elliptical shape, the resist film **33** is exposed using a photomask **37** having a rectangular pattern as the polygonal pattern **37b**, as illustrated in FIG. **36**. In a case where exposure is performed using a photomask **37** having the polygonal pattern **37b** (the rectangular pattern) illustrated in FIG. **36**, an elliptical pattern in which corner parts of the polygonal pattern **37b** (the rectangular pattern) are rounded is transferred to the resist film **33** as illustrated in FIG. **34A**. Here, a guaranteed minimum line width of the opening **34** that can be formed on the resist film **33** is a line of 35 nm and a hole of 62.5 nm in a case where immersion is used. Therefore, as illustrated in FIGS. **34B** and **33B**, for example, when an elliptical shape or a rectangular shape is used as the shape of the opening **34** of the resist film **33**, a maximum width of the groove **39** can be narrowed as compared with a case of using a

circular shape.

(150) (4) Furthermore, in the second embodiment, an example has been described in which the plurality of openings **45** is aligned in one line so as to surround the pixel region **3** in the etching mask (the resist film **33**), but other configurations may also be adopted. For example, as illustrated in FIGS. **37B**, **38B**, and **39B**, the etching mask may have a configuration in which a plurality of lines in which the plurality of openings **45** is aligned so as to surround the pixel region **3** is aligned in a width direction of the groove **39**, when viewed from the back surface S3 side (one surface side) of the substrate **2**.

(151) In FIG. **37B**, a case where shapes of the openings **45** are all the same is exemplified. That is, the same shape (for example, an elliptical shape) is used as the shape of each of the plurality of openings **45**. In a case where an elliptical shape is used as the shape of the opening **45** of the resist film **33**, for example, as illustrated in FIG. **37A**, exposure is performed using a photomask **37** having a rectangular pattern as the polygonal pattern **37b**. Furthermore, in FIGS. **38B** and **39B**, a case is exemplified in which the shapes of the openings **45** are different for every line. That is, two or more types of shapes (for example, elliptical and circular shapes) are used as the shape of each of the plurality of openings **45**. In a case where two types of elliptical and circular shapes are used as the shapes of the openings **45** of the resist film **33**, for example, as illustrated in FIGS. **38A** and **39A**, the resist film **33** is exposed using a photomask **37** having two types of a rectangular pattern and a square pattern as the polygonal pattern **37b**.

(152) Furthermore, in FIGS. **37B** and **38B**, a case is exemplified in which the openings **45** in one line and the openings **45** in another line adjacent to the one line are disposed to be shifted by a half pitch in the direction in which the groove **39** extends. Furthermore, in FIG. **39B**, a case is exemplified in which the openings **45** in one line and the openings **45** in another line adjacent to the one line are disposed without being shifted in the direction in which the groove **39** extends.

(153) Then, by performing isotropic etching in the step of forming the groove **39** by using the etching mask (the resist film **33**) illustrated in FIGS. **37B**, **38B**, and **39B**, the openings **46** in one line and the openings **46** in another line adjacent to the one line are connected as illustrated in FIGS. **37C**, **38C**, and **39C**. As a result, the shape of the side wall surface of the groove **39** can be made complicated.

3. Third Embodiment: Example of Application to Electronic Device

(154) The technology according to the present disclosure (the present technology) may be applied to various electronic devices such as, for example, an imaging device such as a digital still camera or a digital video camera, a mobile phone having an imaging function, or other devices having an imaging function.

(155) FIG. **40** is a diagram illustrating an example of a schematic configuration of an electronic device (for example, a camera) to which the technology according to the present disclosure (the present technology) can be applied.

(156) As illustrated in FIG. **40**, an electronic device **100** includes a solid-state imaging device **101**, an optical lens **102**, a shutter device **103**, a drive circuit **104**, and a signal processing circuit **105**.

(157) The optical lens **102** forms an image of image light (incident light **106**) from a subject on an imaging surface of the solid-state imaging device **101**. As a result, signal charges are accumulated in the solid-state imaging device **101** over a certain period. The shutter device **103** controls a light irradiation period and a light shielding period for the solid-state imaging device **101**. The drive circuit **104** supplies a drive signal for controlling a transfer operation of the solid-state imaging device **101** and a shutter operation of the shutter device **103**. A signal of the solid-state imaging device **101** is transferred by a drive signal (a timing signal) supplied from the drive circuit **104**. The signal processing circuit **105** performs various types of signal processing on a signal (a pixel signal) outputted from the solid-state imaging device **101**. A video signal subjected to the signal processing is stored in a storage medium such as a memory or outputted to a monitor.

(158) Note that the electronic device **100** to which the solid-state imaging device **1** can be applied

is not limited to a camera, and the solid-state imaging device **1** can also be applied to other electronic devices. For example, the solid-state imaging device **1** may be applied to an imaging device such as a camera module for a mobile device such as a mobile phone or a tablet terminal. (159) An example of the electronic device to which the technology according to the present disclosure can be applied has been described above. The technology according to the present disclosure can be applied to the solid-state imaging device **101** among the above-described configurations. Specifically, the solid-state imaging device **1** according to the first embodiment, the solid-state imaging device **1** according to the second embodiment, and the solid-state imaging device **1** according to the modifications thereof can be applied to the solid-state imaging device **101**. By applying the technology according to the present disclosure to the solid-state imaging device **101**, a better captured image can be obtained.

(160) Note that the present technology can have the following configurations.

(161) (1)

(162) A method for manufacturing a light detection device, the method including: performing, on a substrate in which a plurality of photoelectric conversion units is formed, anisotropic etching from one surface side of the substrate to form a plurality of openings arranged in a lattice shape at predetermined intervals on the substrate to surround each of the photoelectric conversion units; and then performing isotropic etching from one surface side of the substrate to connect the openings adjacent to each other to form a trench part having a lattice shape on the substrate to surround each of the photoelectric conversion units.

(2)

(163) The method for manufacturing the light detection device according to (2) above, in which the anisotropic etching and the isotropic etching are performed using an etching mask that is formed on one surface of the substrate, the etching mask having a plurality of openings arranged in a lattice shape at predetermined intervals to surround each of the photoelectric conversion units when viewed from the one surface side of the substrate.

(3)

(164) The method for manufacturing the light detection device according to (1) or (2) above, in which a shape of an opening of the etching mask in a cross section parallel to a light incident surface of the substrate is a circular shape, an elliptical shape, or a polygonal shape.

(4)

(165) The method for manufacturing the light detection device according to any one of (1) to (3) above, the method further including: forming a plurality of second openings arranged at predetermined intervals to surround a pixel region having a plurality of the photoelectric conversion units, simultaneously with formation of first openings that are a plurality of openings arranged to surround each of the photoelectric conversion units on the substrate by the anisotropic etching; and connecting the second openings adjacent to each other simultaneously with formation of the trench part on the substrate by connecting the first openings adjacent to each other by the isotropic etching, to form a groove to surround the pixel region.

(5)

(166) The method for manufacturing the light detection device according to (4) above, in which the anisotropic etching and the isotropic etching are performed using an etching mask that is formed on one surface of the substrate, the etching mask including: a plurality of openings arranged at predetermined intervals to surround the pixel region when viewed from the one surface side of the substrate; and a plurality of openings arranged in a lattice shape at predetermined intervals to surround each of the pixel region.

(6)

(167) The method for manufacturing the light detection device according to (5) above, in which a shape of an opening of the etching mask is a circular shape, an elliptical shape, or a polygonal shape.

(7)
(168) The method for manufacturing the light detection device according to (5) above, in which in the etching mask, when viewed from one surface side of the substrate, a plurality of lines in which a plurality of openings is aligned to surround the pixel region is aligned in a width direction of the groove.

(8)
(169) The method for manufacturing the light detection device according to (7) above, in which as individual shapes of the plurality of openings of the etching mask, shapes identical to one another are used.

(9)
(170) The method for manufacturing the light detection device according to any one of (7) above, in which two or more kinds of shapes are used as each shape of the plurality of openings of the etching mask.

(10)
(171) The method for manufacturing the light detection device according to any one of (1) to (3) above, further including: in a step different from a step of forming the trench part, performing anisotropic etching on the substrate from one surface side of the substrate to form a plurality of openings arranged at predetermined intervals on the substrate to surround a pixel region having a plurality of the photoelectric conversion units, and then performing isotropic etching from one surface side of the substrate to connect the second openings adjacent to each other, to form a groove to surround the pixel region.

(11)
(172) The method for manufacturing the light detection device according to any one of (1) to (19) above, in which the anisotropic etching is performed using a Bosch process.

(12)
(173) A light detection device including: a substrate; a plurality of photoelectric conversion units formed in the substrate; and a trench part having a lattice shape and formed on the substrate to surround each of the photoelectric conversion units, in which a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a longitudinal direction of a linear trench part constituting the trench part.

(13)
(174) The light detection device according to (12) above, in which the concavo-convex shape is a wave shape in which arcs having a radius of 50 nm or more and 150 nm or less are continuous.

(14)
(175) The light detection device according to (12) or (13) above, in which a cross-sectional shape of a side wall surface of the trench part in a cross section perpendicular to the light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a depth direction of the trench part.

(15)
(176) The light detection device according to any one of (12) to (14) above, in which the light detection device is a CMOS image sensor in which a color filter layer having a plurality of color filters and a microlens array having a plurality of microlenses are layered in this order on the light incident surface side of the substrate.

(16)
(177) The light detection device according to (15) above, in which the microlens is formed for each photoelectric conversion unit group including the photoelectric conversion units disposed in a 2×2 matrix shape.

(17)
(178) The light detection device according to any one of (12) to (14) above, in which the light

detection device is a distance measuring sensor in which a microlens array including a plurality of microlenses is layered on the light incident surface side of the substrate.

(18)

(179) The light detection device according to any one of (12) to (17) above, further including: a groove formed to surround a pixel region including the plurality of photoelectric conversion units, the groove being formed between the pixel region and a blade region surrounding the pixel region in the substrate, in which a cross-sectional shape of a side wall surface of the groove in a cross section parallel to the light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a direction in which the groove extends.

(19)

(180) The light detection device according to (18) above, in which a maximum width of the groove is 500 nm or less.

(20)

(181) An electronic device including: a light detection device including a substrate, a plurality of photoelectric conversion units formed in the substrate, and a trench part having a lattice shape and formed on the substrate to surround each of the photoelectric conversion units, in which a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a longitudinal direction of a linear trench part constituting the trench part; an optical lens configured to form an image of image light from a subject on an imaging surface of the light detection device; and a signal processing circuit configured to perform signal processing on a signal outputted from the light detection device.

(21)

(182) A method for manufacturing a light detection device, the method including: performing anisotropic etching on a substrate in which a plurality of photoelectric conversion units is formed from one surface side of the substrate to form a plurality of openings arranged at predetermined intervals from one another on the substrate, and then performing isotropic etching from one surface side of the substrate to connect the openings adjacent to each other, to form a groove on the substrate along an arrangement direction of the openings.

(22)

(183) The method for manufacturing the light detection device according to (21) above, in which each of a plurality of the openings is an opening arranged at a predetermined interval from one another to surround each of the photoelectric conversion units, and the groove is a trench part disposed to surround each of the photoelectric conversion units.

(23)

(184) The light detection device according to (21) above, in which each of a plurality of the openings is an opening arranged at a predetermined interval from one another to surround a pixel region including the plurality of photoelectric conversion units, and the groove is a groove disposed to surround the pixel region.

(24)

(185) A light detection device including: a substrate; a plurality of photoelectric conversion units formed in the substrate; and a trench part formed on the substrate to surround each of the photoelectric conversion units, in which a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a direction in which the trench part extends.

(25)

(186) A light detection device including: a substrate; a plurality of photoelectric conversion units formed in the substrate; and a groove formed to surround a pixel region including the plurality of photoelectric conversion units, the groove being formed between the pixel region and a blade region surrounding the pixel region in the substrate, in which a cross-sectional shape of a side wall

surface of the groove in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a direction in which the groove extends, and a maximum width of the groove is 500 nm or less.

(26)

(187) An electronic device including: a light detection device including a substrate, a plurality of photoelectric conversion units formed in the substrate, and a light detection device including a groove formed to surround a pixel region including the plurality of photoelectric conversion units, the groove being formed between the pixel region and a blade region surrounding the pixel region in the substrate, in which a cross-sectional shape of a side wall surface of the groove in a cross section parallel to a light incident surface of the substrate is a concavo-convex shape in which concavity and convexity are repeated along a direction in which the groove extends, and a maximum width of the groove is 500 nm or less.

REFERENCE SIGNS LIST

(188) **1** Solid-state imaging device **2** Substrate **3** Pixel region **4** Vertical drive circuit **5** Column signal processing circuit **6** Horizontal drive circuit unit **7** Output circuit **8** Control circuit **9** Pixel **10** Pixel drive wiring **11** Vertical signal line **12** Horizontal signal line **13** Fixed charge film **14** Insulating film **15** Light shielding film **16** Planarization film **17** Light receiving layer **18** Color filter layer **19** Microlens array **20** Light condensing layer **21** Wiring layer **22** Supporting substrate **23** Photoelectric conversion unit **23a** Corner part **24** Pixel separation part **25** Trench part **26** Vertical hole **27** Intersection part **28** Concave part **29** Color filter **30** Microlens **31** Interlayer insulating film **32** Wiring **33** Resist film **34** Opening **35** Opening **36** Photoelectric conversion unit group **37** Photomask **38** Chip **39** Groove **40** Sensor substrate **41** Logic substrate **42** Blade region **43** Vertical hole **44** Concave part **45** Opening **46** Opening **47** Resist film **100** Electronic device **101** Solid-state imaging device **102** Optical lens **103** Shutter device **104** Drive circuit **105** Signal processing circuit **106** Incident light

Claims

1. A method for manufacturing a light detection device, the method comprising: performing, on a substrate in which a plurality of photoelectric conversion units is formed, anisotropic etching from one surface side of the substrate to form a plurality of openings arranged in a lattice shape at predetermined intervals on the substrate to surround each of the photoelectric conversion units; and then performing isotropic etching from the one surface side of the substrate to connect the plurality of openings adjacent to each other to form a trench part having a lattice shape on the substrate to surround each of the photoelectric conversion units.
2. The method for manufacturing the light detection device according to claim 1, wherein the anisotropic etching and the isotropic etching are performed using an etching mask that is formed on the one surface side of the substrate, the etching mask having a plurality of openings arranged in a lattice shape at predetermined intervals to surround each of the photoelectric conversion units when viewed from the one surface side of the substrate.
3. The method for manufacturing the light detection device according to claim 2, wherein a shape of an opening of the plurality of openings of the etching mask in a cross section parallel to a light incident surface of the substrate includes a circular shape, an elliptical shape, or a polygonal shape.
4. The method for manufacturing the light detection device according to claim 1, further comprising: forming a plurality of second openings arranged at predetermined intervals to surround a pixel region having a plurality of the photoelectric conversion units, simultaneously with formation of first openings that are the plurality of openings arranged to surround each of the photoelectric conversion units on the substrate by the anisotropic etching; and connecting the plurality of second openings adjacent to each other simultaneously with formation of the trench part on the substrate by connecting the first openings adjacent to each other by the isotropic

etching, to form a groove to surround the pixel region.

5. The method for manufacturing the light detection device according to claim 4, wherein the anisotropic etching and the isotropic etching are performed using an etching mask that is formed on the one surface side of the substrate, the etching mask including: a first plurality of openings arranged at predetermined intervals to surround the pixel region when viewed from the one surface side of the substrate; and a second plurality of openings arranged in a lattice shape at predetermined intervals to surround each of the pixel region.

6. The method for manufacturing the light detection device according to claim 5, wherein a shape of an opening of the first plurality of openings or the second plurality of openings of the etching mask includes a circular shape, an elliptical shape, or a polygonal shape.

7. The method for manufacturing the light detection device according to claim 5, wherein in the etching mask, when viewed from the one surface side of the substrate, a plurality of lines in which the second plurality of openings is aligned to surround the pixel region is aligned in a width direction of the groove.

8. The method for manufacturing the light detection device according to claim 7, wherein as individual shapes of the first plurality of openings or the second plurality of openings of the etching mask, shapes identical to one another are used.

9. The method for manufacturing the light detection device according to claim 7, wherein two or more kinds of shapes are used as each shape of the first plurality of openings or the second plurality of openings of the etching mask.

10. The method for manufacturing the light detection device according to claim 1, further comprising: in a step different from a step of forming the trench part, performing anisotropic etching on the substrate from the one surface side of the substrate to form a plurality of second openings arranged at predetermined intervals on the substrate to surround a pixel region having a plurality of the photoelectric conversion units, and then performing isotropic etching from the one surface side of the substrate to connect the plurality of second openings adjacent to each other, to form a groove to surround the pixel region.

11. The method for manufacturing the light detection device according to claim 1, wherein the anisotropic etching is performed using a Bosch process.

12. A light detection device comprising: a substrate; a plurality of photoelectric conversion units formed in the substrate; and a trench part having a lattice shape and formed on the substrate to surround each of the plurality of photoelectric conversion units, wherein a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate includes a concavo-convex shape in which concavity and convexity are repeated along a longitudinal direction of a linear trench part constituting the trench part.

13. The light detection device according to claim 12, wherein the concavo-convex shape includes a wave shape in which arcs having a radius of 50 nm or more and 150 nm or less are continuous.

14. The light detection device according to claim 12, wherein a cross-sectional shape of a side wall surface of the trench part in a cross section perpendicular to the light incident surface of the substrate includes a concavo-convex shape in which concavity and convexity are repeated along a depth direction of the trench part.

15. The light detection device according to claim 12, wherein the light detection device further includes a CMOS image sensor in which a color filter layer having a plurality of color filters and a microlens array having a plurality of microlenses are layered in this order on a light incident surface side of the substrate.

16. The light detection device according to claim 15, wherein the microlens array includes a microlens formed for each photoelectric conversion unit group including the plurality of photoelectric conversion units disposed in a 2×2 matrix shape.

17. The light detection device according to claim 12, wherein the light detection device further includes a distance measuring sensor in which a microlens array including a plurality of

microlenses is layered on a light incident surface side of the substrate.

18. The light detection device according to claim 12, further comprising: a groove formed to surround a pixel region including the plurality of photoelectric conversion units, the groove being formed between the pixel region and a blade region surrounding the pixel region in the substrate, wherein a cross-sectional shape of a side wall surface of the groove in a cross section parallel to the light incident surface of the substrate includes a concavo-convex shape in which concavity and convexity are repeated along a direction in which the groove extends.

19. The light detection device according to claim 18, wherein a maximum width of the groove is 500 nm or less.

20. An electronic device, comprising: a light detection device including a substrate, a plurality of photoelectric conversion units formed in the substrate, and a trench part having a lattice shape and formed on the substrate to surround each of the plurality of photoelectric conversion units, wherein a cross-sectional shape of a side wall surface of the trench part in a cross section parallel to a light incident surface of the substrate includes a concavo-convex shape in which concavity and convexity are repeated along a longitudinal direction of a linear trench part constituting the trench part; an optical lens configured to form an image of image light from a subject on an imaging surface of the light detection device; and a signal processing circuit configured to perform signal processing on a signal outputted from the light detection device.
